

Group Comparisons with Analysis of Variance

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Table of contents

1	Comparison Questions and Designs	4
1.1	Outcome, factors, cells and units	4
1.2	Two studies, two questions	5
1.3	Design determines the relevant comparison	6
2	One-Way ANOVA	7
2.1	Inspect grouped observations before reducing them	7
2.2	Mean model and the partition of variation	7
2.3	Degrees of freedom, mean squares and the F test	9
2.4	Two groups, direction and magnitude	13
3	Contrasts and Multiple Comparisons	14
3.1	State the comparison through its weights	14
3.2	Calculate the uncertainty for that comparison	15
3.3	Families of comparisons and the familywise error rate	16
4	Multi-Way (Factorial) ANOVA	19
4.1	What interaction means, and why study two factors at once	19
4.2	Main effects and interaction through graphs	20
4.3	Cells, marginal means and the full data	21
4.4	Additivity and the full model	22
4.5	Simple effects and a selected interaction contrast	27
4.6	Give a connected interpretation	28
4.7	Three and four factors	29
5	Checking the Conditions	31
5.1	What the design and the data can show	31
5.2	When spreads differ: Welch's one-way ANOVA	33

6 Reading Bridge	35
Key terms	35
Further reading	37
Exercises	37
Exercise 1: Read grouped observations (Foundational)	37
Exercise 2: Read the one-way table (Foundational)	37
Exercise 3: Define and estimate contrasts (Foundational)	38
Exercise 4: Interpret comparison families (Integrative)	39
Exercise 5: Read the factorial pattern (Foundational)	40
Exercise 6: Distinguish joint and selected results (Integrative)	40
Exercise 7: Repair a claim about teaching (Integrative)	41
Exercise 8: Match the analysis to the design (Integrative)	42
Appendix A: Deriving the Matched Pooled-t Identity	42
Appendix B: Weighting and Effect-Size Reference	43
B.1 Unequal-cell weights	43
B.2 Effect-size denominators	43
References	44

Chapter overview

This chapter compares means by examining variation. Two invented studies provide the observations: a one-way experiment on background sound while studying carries the first analyses, and a two-factor experiment crossing feedback with practice schedule carries the factorial analysis, which asks how a comparison changes across practice schedules. We follow the one-way ANOVA from individual scores to its F distribution, distinguish its omnibus question from a particular contrast, and explain how the comparison family determines an interval's coverage. Cell means and simple effects then reveal what main effects can conceal, and a closing look at three and four factors shows how a two-way pattern can itself change. By the end, you will be able to read an ANOVA table and the F distribution behind it, interpret a contrast and its uncertainty within a family of comparisons, and connect a factorial pattern to the design and model conditions that support its interpretation.

Two **invented studies** supply the observations. In the first, thirty first-year university students each read the same passage for 20 minutes and then answer a 20-item reading-comprehension quiz, scored from 0 to 20. Each student reads in an individual session and is randomly assigned to one of three background-sound conditions, ten to each: *Silence*, *Instrumental music*, or *Music with lyrics* playing during the reading. In the second, twenty-four students study the same fractions material and complete a common 20-item quiz. Random allocation places four different students in each combination of three feedback conditions and two practice schedules. *Rereading* offers the worked material again; *Answer feedback* supplies the correct answer after an attempt; *Explanatory feedback* supplies the answer with a worked explanation. *Massed* practice occurs in one session and *Spaced* practice in two shorter sessions with the same total practice time. Everyone completes the quiz at the same specified follow-up point after their final practice. These design details and all scores are invented for teaching.

Our first question concerns the background-sound study: how do the three sound-condition means compare? Chapter 3 built, step by step, the tools for one comparison: an estimate and its standard error, a confidence interval, a hypothesis test with its p-value, and the t tests for one sample, paired samples and two independent samples. For two independent groups sharing one spread, the equal-variance t test pooled the two group variances. We extend that reasoning to several groups, keeping the difference between an overall pattern and particular comparisons visible.

Our later question concerns the feedback-by-practice study: how does the feedback comparison depend on practice schedule? The number of factors, not the number of groups, separates the two analyses. The first study has one factor with three levels; the second has two crossed factors forming six cells.

1 Comparison Questions and Designs

1.1 Outcome, factors, cells and units

Both outcomes count correct answers. A comprehension score is the number of reading-comprehension items answered correctly, and a quiz score is the number of fractions items answered correctly; each runs from 0 to 20, and higher values mean more correct answers. A four-point difference therefore means four more correct answers on average; it does not by itself describe every aspect of reading comprehension or of learning fractions. The outcome and its scale give a numerical comparison its substantive meaning.

A **factor** in analysis of variance is a categorical variable defining groups or conditions. Its categories are **levels**. Background sound has three levels. In the second study, feedback has three levels and practice schedule has two. This use of *factor* differs from a latent factor used to represent shared indicator variation in a measurement model.

Table 1

Invented reading-comprehension scores, ten different students per sound condition, on a 0–20 scale.

Sound condition	Comprehension scores of ten different students
Silence	8, 9, 9, 11, 12, 13, 13, 14, 15, 16
Instrumental music	6, 8, 8, 9, 9, 11, 11, 12, 12, 14
Music with lyrics	4, 5, 6, 7, 8, 8, 9, 10, 11, 12

One factor with three levels gives a **one-way** layout of three groups. Each score in Table 1 belongs to a different student: the ten scores of one sound condition describe ten different students, not repeated measurements, and the student is both the stipulated assignment unit and the measurement unit. The scores are fixed teaching values. They place the three group means two points apart with similar, unequal spreads, so every calculation reproduces by hand, but they do not establish a Normal model, equal variances or a real effect of background sound.

The complete course data file, `background_sound.csv`, represents the same table in one row per student, with columns `student`, `sound` and `score`. Students are numbered S01–S30, ten to each sound condition in the order Silence, Instrumental music, Music with lyrics. Silence is the reference condition for the comparisons in Section 3.

A **cell** contains cases sharing one level of each factor. Answer feedback with Spaced practice is one cell. Crossing three feedback levels with two practice levels gives six cells, a **two-way** layout. Four students per cell makes this layout **balanced**, as ten students per condition balance the one-way layout. One-way and two-way refer to the number of factors, not to the

number of levels. A one-way analysis can compare three, four or more groups, and a third or fourth crossed factor gives a three-way or four-way layout (Section 4.7).

Table 2

Invented fractions-quiz scores, four different students per cell, on a 0–20 scale.

Feedback condition	Massed practice	Spaced practice
Rereading	4, 5, 7, 8	4, 5, 7, 8
Answer feedback	6, 7, 9, 10	10, 11, 13, 14
Explanatory feedback	8, 9, 11, 12	16, 17, 19, 20

Each score in Table 2 belongs to a different student. The student is both the stipulated assignment unit and the measurement unit. The four scores in a cell are not repeated measurements on one student; nor are the Massed and Spaced scores paired. The repeated within-cell pattern deliberately simplifies the arithmetic. It is a teaching construction, not evidence of Normal errors, equal variances or an educational effect.

The complete course data file, `feedback_practice.csv`, represents the same table in one row per student, with columns `student`, `feedback`, `practice` and `score`. Students are numbered 1–24, ordered by feedback, then practice, then student within each cell. The printed scores in the two tables are sufficient for every calculation in this chapter.

1.2 Two studies, two questions

For the first analysis, the background-sound study has three groups defined by one factor. Their model means, written μ_S , μ_I and μ_L , refer to Silence, Instrumental music and Music with lyrics. For example, $\mu_L - \mu_S$ states a direction and an outcome scale: how many more or fewer comprehension items students answer correctly, on average, with lyrics playing than in silence. The observed difference estimates this particular contrast under the model introduced below.

The feedback-by-practice study crosses two factors, so its six cells support a second kind of question: does the feedback comparison change with practice schedule? Section 4 answers it with all 24 students.

Table 3
Two questions, each asked of its own invented study.

Analysis	Records used	Question addressed
One-way	Background-sound study: 30 students in three sound conditions, ten per condition	Do the mean comprehension scores differ across the three sound conditions?
Two-way	Feedback-by-practice study: 24 students in six feedback-by-practice cells, four per cell	What are the average factor comparisons, and how do comparisons change across the other factor?

A two-factor ANOVA and the two-way frequency table of Chapter 3, Section 6, have different outcomes. ANOVA compares a numerical score across factor combinations. A frequency table counts cases in combinations of categories. The word *two-way* describes a layout; it does not identify a shared statistical test.

1.3 Design determines the relevant comparison

Random allocation is stipulated in each invented story. If implemented in a real study, that design could support a causal comparison among the participating students, subject to adherence, relevant interference and missing outcomes. It would not make the participants a random sample of all students. The data table alone establishes neither implementation of assignment nor representativeness.

Different designs can produce superficially similar tables. If students who heard one sound condition read together in one room, they would share that session's influences, such as noise from a corridor or a disruption partway through the reading. If whole classes receive conditions, students within the same class may share influences and the class may be the assignment unit. If each student is measured twice, the two outcomes belong to a pair. For a real study, identify the allocation, any repeated measurements, shared sessions or classroom clustering, and any losses before choosing the error term. An uncertainty formula must reflect these connections rather than treating every row as an independent assignment.

Blocking makes condition comparisons within groups of similar units. Chapter 3, Section 2.3, illustrates it with paired schools: in each of twelve schools one class is assigned to each format, so the uncertainty follows twelve independent within-school comparisons rather than 480 students. The three sound conditions and the six feedback-by-practice cells in this chapter instead contain distinct, individually assigned students. Sampling strata, which organize selection into a sample, and allocation blocks, which organize treatment comparisons, have different roles.

2 One-Way ANOVA

2.1 Inspect grouped observations before reducing them

The three sound conditions have means 12, 10 and 8 comprehension points. Each has ten scores, and the sample SDs, 2.7080, 2.4037 and 2.5820, are similar. The ranges overlap widely: some students who read with lyrics playing, scoring 11 or 12, outsourced some who read in silence, scoring 8 or 9. A difference between means summarizes group distributions; it does not say that every student in one group outperformed every student in another.

Table 4

Descriptions of the thirty background-sound scores.

Sound condition	n	Mean	SD	Range
Silence	10	12	2.7080	8–16
Instrumental music	10	10	2.4037	6–14
Music with lyrics	10	8	2.5820	4–12

Figure 1 draws the same thirty scores in its left panel: each box shows a group’s median and middle half, and the points keep every student visible, which a bar chart of the three means would hide. That within-group variation matters to inference, because the same distance between group means can be more or less convincing depending on how widely individual scores vary within groups. The right panel makes the point by construction: it doubles every score’s distance from its group mean, so the group means and counts stay fixed while the spread doubles.

Doubling every distance from the group mean multiplies its square by four, so the within-group squared variation quadruples while the separation of the means is unchanged. The group SDs double, from 2.7080, 2.4037 and 2.5820 to 5.4160, 4.8074 and 5.1640. That change alone moves the F statistic printed beneath each panel from 6.0674 to 1.5169, one quarter of its value, and the p-value from .0067 to .2375: the same two-point steps between the means stand out against narrow within-group spread and are lost in wide spread. ANOVA compares these two kinds of variation after putting them on comparable scales (Çetinkaya-Rundel & Hardin, 2024; Kutner et al., 2005). Section 2.3 develops the F statistic and its p-value, and returns to this figure with the two other quantities that move F.

2.2 Mean model and the partition of variation

Write Y_{ij} for the random score of student i in sound condition j . Here j indexes a group, rather than a data-file column. The one-way model is

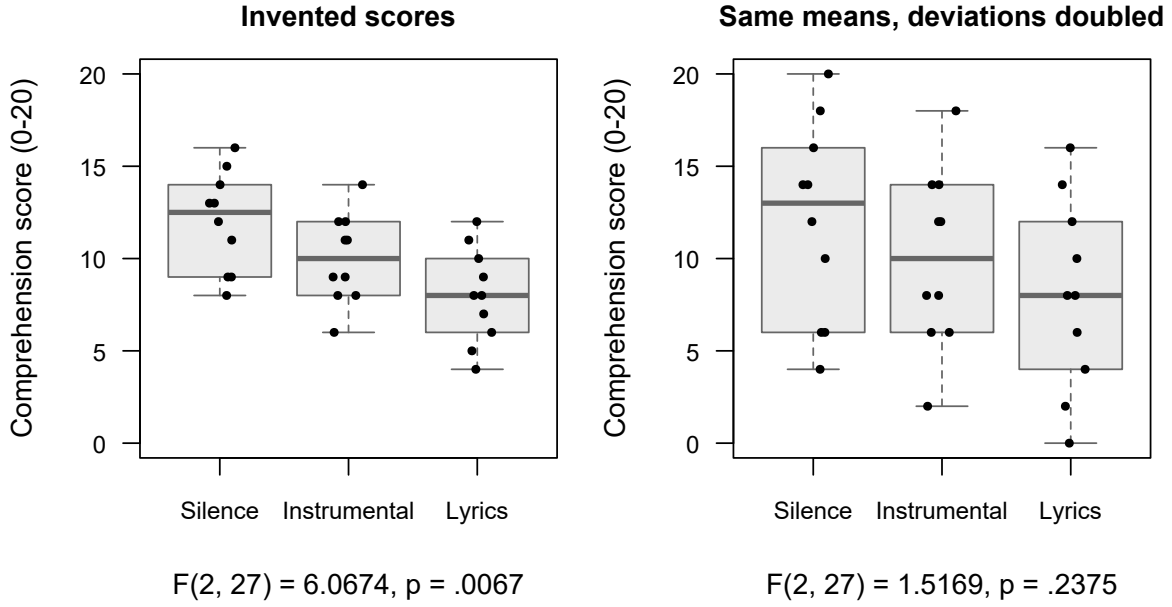


Figure 1: The idea of ANOVA: the same group means with different within-group spread. Left: the thirty invented scores. Right: a constructed version in which every score's distance from its group mean is doubled, so the group means and counts are unchanged, although the medians that the boxes mark can shift. Boxes show each group's median and middle half; each point is a student, jittered sideways so that equal scores stay visible. Each panel's F statistic and p-value appear beneath it.

$$Y_{ij} = \mu_j + \varepsilon_{ij}.$$

The unknown μ_j is the group mean in the model, and ε_{ij} is that student's unknown departure from it. After fitting the group means, the **fitted value** is $\hat{y}_{ij} = \bar{y}_j$, and the **residual** is $e_{ij} = y_{ij} - \bar{y}_j$. A residual is calculated from observed scores and fitted means; a model error is a departure from an unknown model mean. This distinction will recur in regression.

The observed grand mean is $\bar{y} = 10$, the average of the three group means because the groups are equally sized. Every score's distance from the grand mean splits into two parts, the group's departure from the grand mean and the student's residual:

$$y_{ij} - \bar{y} = (\bar{y}_j - \bar{y}) + (y_{ij} - \bar{y}_j).$$

For the Music-with-lyrics student who scored 4, $4 - 10 = (8 - 10) + (4 - 8)$: the group mean lies two points below the grand mean, and this student another four points below the group

mean. The first part is shared by every student in the group; the second differs from student to student.

Squaring the parts and adding over all thirty students turns the split into a partition of the total squared variation, $SS_T = SS_B + SS_W$, where

$$\begin{aligned} SS_T &= \sum_{j=1}^J \sum_{i=1}^{n_j} (y_{ij} - \bar{y})^2, \\ SS_B &= \sum_{j=1}^J n_j (\bar{y}_j - \bar{y})^2, \\ SS_W &= \sum_{j=1}^J \sum_{i=1}^{n_j} (y_{ij} - \bar{y}_j)^2. \end{aligned}$$

The **total sum of squares** SS_T measures variation around the one grand mean. The **between-group sum of squares** SS_B counts each group's departure once for every student in that group, so it grows as the group means separate. The **within-group sum of squares** SS_W adds the squared residuals around the group means. The partition holds because, within each group, the cross-products $2(\bar{y}_j - \bar{y})(y_{ij} - \bar{y}_j)$ sum to zero: the group's departure is the same for every student, and the residuals of a group sum to zero. For the background-sound study, $SS_B = 80$, $SS_W = 178$ and $SS_T = 258$, in squared comprehension points. The Instrumental-music group adds nothing to SS_B , because its mean equals the grand mean, although its scores still vary. The partition is arithmetic; it needs no Normality or independence assumption. Those conditions enter when the partition is used for inference, in Section 2.3.

2.3 Degrees of freedom, mean squares and the F test

A sum of squares grows with both the size of the departures and the number of them, so each is divided by its **degrees of freedom** (df), the number of departures that can vary freely, before the two sources are compared. The F statistic has two kinds of df.

- The **between-groups df**, $J - 1 = 2$, are the numerator df. The three group means can take any values, but once the grand mean is fixed their departures from it, weighted by group size, sum to zero, so only two of them are free.
- The **within-groups df**, $N - J = 27$, are the denominator df. The ten residuals of each group sum to zero, so nine of them are free in each group, and three groups give 27.

The two kinds add to the total df, $N - 1 = 29$, because the thirty departures from the grand mean sum to zero. Dividing each sum of squares by its df gives a **mean square**,

$$MS_B = \frac{SS_B}{J - 1}, \quad MS_W = \frac{SS_W}{N - J}, \quad F = \frac{MS_B}{MS_W}.$$

Table 5 gathers the parts.

Table 5

One-way ANOVA for the 30 students of the background-sound study, with what each df counts.

Source	SS	df	What the df count	MS	F	p
Sound condition	80	2	Group means free to vary, $J - 1$	40.0000	6.0674	.0067
Within groups	178	27	Residuals free to vary after fitting three means, $N - J$	6.5926		
Total	258	29	Departures from the grand mean free to vary, $N - 1$			

Read the table as one sentence about the observations, $\text{Data} = \text{fit} + \text{residual}$. The sound-condition row is the fit, what the three group means account for; the within-groups row is the residual, what is left around them; and the total row is the data, measured around the grand mean. The sums of squares add across the rows, $80 + 178 = 258$, and so do the df, $2 + 27 = 29$, which makes a quick check of any printed table. $MS_W = 6.5926$ estimates the common error variance of the model; it is smaller than the total sample variance, $258/29 = 8.8966$, because it leaves out the separation of the means. Statistical software fits the same model from the thirty records and often prints a longer layout, with rows such as an intercept and an uncorrected total that serve the fitting rather than the comparison question; the Computer Lab reads one such layout row by row.

The omnibus null hypothesis is $H_0 : \mu_S = \mu_I = \mu_L$. The alternative says that at least two of these means differ. Judging whether 6.0674 is large needs the distribution that the statistic would follow if the null were true.

Three conditions support this reference distribution, stated first in familiar words. **Independent observations:** one student's departure from a group mean carries no information about another student's departure. Here that rests on the design, with students individually assigned, each reading in an individual session and scored once; no display of the scores can establish it. **Approximately Normal residuals:** the departures around each group mean, not the pooled mixture of all thirty scores, have a roughly Normal shape. **Similar variances within groups:** one error variance σ^2 describes every group, which is what allows MS_W to pool the three group variances. In model shorthand, the errors ε_{ij} are independent Normal with mean zero and common variance σ^2 .

Under the null, both mean squares estimate the same error variance, so their ratio tends to lie near 1; the numerator also grows with the separation of the model means, so a large ratio is evidence against equality. Under the three conditions and the null, F follows an **F distribution**, one of a family of right-skewed distributions on positive values, each indexed by its two df, numerator first: here $F(2, 27)$. The left panel of Figure 2 shows three members

of the family and marks each one's 5% point, the value beyond which 5% of the distribution lies. With few denominator df, as in $F(2, 8)$, the error variance is poorly estimated and large ratios occur more often by chance: its curve looks much like that of $F(2, 27)$, but its heavier tail moves the 5% point from 3.3541 to 4.4590. With more numerator df, as in $F(5, 27)$, the distribution gathers near 1 and its 5% point falls to 2.5719.

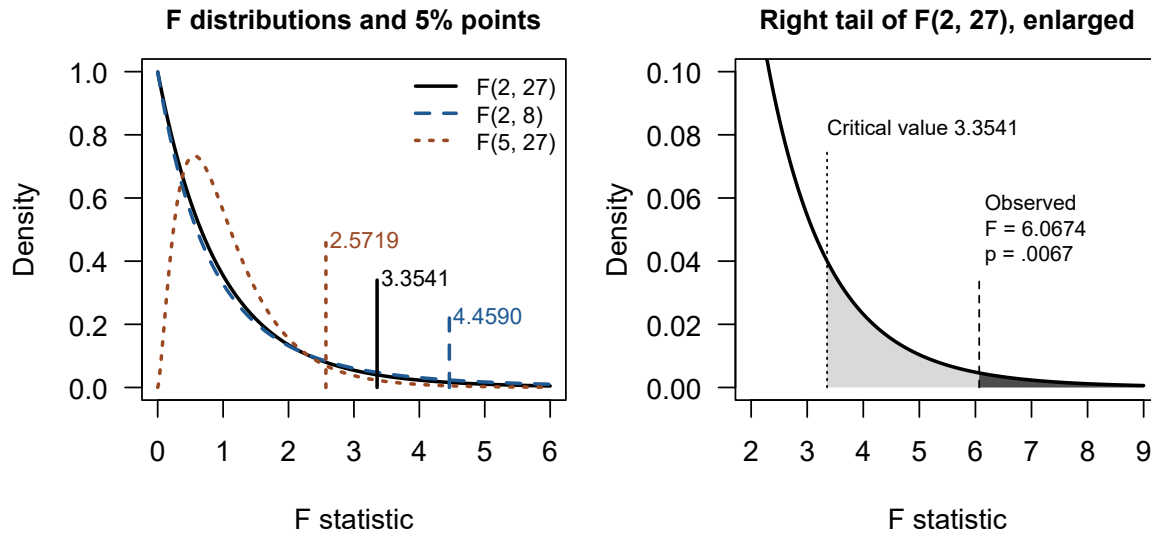


Figure 2: F distributions and the one-way statistic. Left: three F distributions, each indexed by its numerator and denominator df; a vertical mark in each curve's colour and line type, labelled with its value, shows the point beyond which 5% of that distribution lies. Right: the right tail of $F(2, 27)$, enlarged. The shaded areas together are the .05 beyond the critical value 3.3541; the dark area beyond the observed statistic 6.0674 is the p-value, .0067.

The dark area in the right panel of Figure 2, .0067, is read in the same way as the tail areas and p-values of Chapter 3, Section 4. Only the right tail counts, because greater separation of the means produces a larger ratio. Under the null and the model conditions, a ratio at least as large as 6.0674 occurs with that probability. The observed F lies beyond the critical value 3.3541, so at a prespecified .05 significance level the equality null is rejected; comparing F with the critical value and comparing p with .05 give the same decision. This is evidence about the three-mean pattern, not a probability that the null is true and not an identification of which particular comparisons differ.

The box below states, in the four lines used since Chapter 3, what this F test addresses, what it rests on, what the data can check and what the result supports.

💡 Justifying the analysis

- **Target:** whether the model means of the three sound conditions, μ_S , μ_I and μ_L , are all equal for the thirty students of the background-sound study, on the 0–20 comprehension scale; the observed means 12, 10 and 8 estimate them.
- **Conditions:**
 - *Secured by design:* students individually and randomly assigned to the sound conditions, ten to each, as the invented study stipulates, so a difference among the means reads as an effect of background sound among these students; each student read in an individual session and was scored once, so the thirty scores come from independent units. Assignment does not make the students a random sample of a wider population.
 - *Model assumptions:* Normal errors around each group mean with one common variance σ^2 , estimated by $MS_W = 6.5926$ on 27 df.
 - *Choices and defaults:* all thirty students; a .05 significance level fixed in advance; the right tail of $F(2, 27)$, because only greater separation of the means enlarges F ; the classical calibration that pools the group variances; all thirty scores observed, so no rule for missing outcomes is needed.
- **Check:** the group SDs, 2.7080, 2.4037 and 2.5820, with ten students in each group, are similar, the largest 1.1266 times the smallest, and the boxplots show no outlying score. Markedly unequal SDs, above all with unequal group sizes, would call for Welch’s one-way ANOVA (Section 5.2), chosen before the analysis. Ten scores per group say only a little about the error shape, and independence rests on the design, which the scores cannot check.
- **Bound:** $F(2, 27) = 6.0674$, $p = .0067$ is evidence at the .05 level against equal means for these thirty students under the model. It does not identify which means differ or by how much (Section 3), and, with invented scores, says nothing about real studying.

With equal group sizes n , the numerator of F shows directly what the statistic compares:

$$F = \frac{n \sum_{j=1}^J (\bar{y}_j - \bar{y})^2 / (J - 1)}{MS_W}.$$

Three quantities move it: the separation of the group means, the group size n and the within-group spread in MS_W . Table 6 changes one of them at a time, starting from the thirty invented scores; the second row is the right panel of Figure 1.

Table 6

What moves the F statistic: one change at a time to the background-sound study. The last three rows are constructed variations of the invented scores, not further studies; the last has 2 and 57 df, the others 2 and 27.

Version of the study	n	Group means	MS_B	MS_W	F	p
The thirty invented scores	10	12, 10, 8	40	6.5926	6.0674	.0067
Spread doubled	10	12, 10, 8	40	26.3704	1.5169	.2375
Separation halved	10	11, 10, 9	10	6.5926	1.5169	.2375
Group size doubled	20	12, 10, 8	80	6.5926	12.1348	<.0001

Doubling every distance from the group mean multiplies MS_W by four, and halving every mean's distance from the grand mean divides MS_B by four: each change alone quarters F. F therefore reads separation relative to spread, never either alone. Doubling the group size with the same means and SDs leaves every boxplot looking the same, yet doubles MS_B and so F: more students supply more information about the same pattern. Section 2.4 shows that a summary of magnitude, eta squared, barely responds to that third change.

2.4 Two groups, direction and magnitude

ANOVA extends a question about two means to several. For two independent groups sharing one population variance, Chapter 3, Section 5.3, used the **equal-variance two-sample t test**, whose SE, $s_p \sqrt{1/n_1 + 1/n_0}$, is built from the pooled variance s_p^2 of Chapter 3, Section 2.2. The within-group mean square is that pooled variance extended to J groups, each group's sample variance s_j^2 weighted by its own df:

$$MS_W = \frac{SS_W}{N - J} = \frac{\sum_{j=1}^J (n_j - 1) s_j^2}{\sum_{j=1}^J (n_j - 1)}.$$

With equal group sizes, MS_W is the plain average of the group variances, here 7.3333, 5.7778 and 6.6667. With two groups it is Chapter 3's s_p^2 , so the classical one-way F test and the equal-variance t test share one model, one pooled variance and one residual df, and $F(1, N-2) = t_{N-2}^2$. Chapter 3 labels its two groups 1 and 0; this chapter numbers groups $j = 1, \dots, J$, so its two-group case uses 1 and 2.

For Silence and Music with lyrics alone, the pooled variance is 7 on 18 df, and $t(18) = 3.3806$ and $F(1, 18) = 11.4286 = 3.3806^2$ have the same two-sided $p = .0033$. Squaring removes direction, so the signed difference, Silence minus Music with lyrics, 4 points, carries it. This matched fit reads with the *Justifying the analysis* box that Chapter 3 gave the equal-variance t test; only the target changes, to the Silence-minus-Music-with-lyrics difference for these twenty

students, whose SDs, 2.708 and 2.582, are similar. The two-group refit pools 7.0000 on 18 df rather than the three-group $MS_W = 6.5926$ on 27, because it leaves out the Instrumental group, and it answers a narrower question than the three-group F: one pairwise comparison rather than the whole pattern. Appendix A derives the identity, and Section 5.2 returns to groups whose spreads may differ.

Magnitude begins on the comprehension scale. The observed Instrumental-music-minus-Silence difference is -2 points and the Music-with-lyrics-minus-Silence difference is -4 : two and four fewer items answered correctly, on average, than in silence. Whether a difference of this size matters for studying needs a substantive criterion, not only a p-value. One descriptive summary of the overall separation is **eta squared**,

$$\eta^2 = \frac{SS_B}{SS_T} = \frac{80}{258} = .3101 :$$

group means account for .3101 of the observed squared variation in these thirty records. This is a fraction of a sample decomposition, not the proportion of comprehension caused by background sound. Eta squared and F answer different questions: in the last row of Table 6, twenty students per group double F while eta squared barely moves, from .3101 to .2986, because the means account for about the same share of the variation while the larger study gives stronger evidence that they differ. A small p-value therefore does not signal a large difference. Because sample means differ even when model means are equal, eta squared slightly overstates the share in the model; omega squared adjusts for that and gives .2525 here (Appendix B). A standardized mean difference, explained in Chapter 3, Section 7.1, divides a difference by a stated SD; dividing it by an individual-score SD gives a different quantity from dividing it by the SE of that difference.

3 Contrasts and Multiple Comparisons

3.1 State the comparison through its weights

The omnibus question asks whether all three model means agree. A **contrast** states a particular comparison as a weighted combination of means whose weights sum to zero:

$$C = \sum_{j=1}^J c_j \mu_j, \quad \sum_{j=1}^J c_j = 0, \quad \widehat{C} = \sum_{j=1}^J c_j \bar{y}_j.$$

The symbol C here names a contrast, and $\widehat{C}$ its estimate. The weights specify both the comparison and its direction.

The study plan declares two questions, each music condition against Silence: Instrumental music minus Silence and Music with lyrics minus Silence. A negative estimate means fewer

items answered correctly than in silence. A broader question compares the equal average of the two music conditions with Silence. In the declared order Silence, Instrumental music, Music with lyrics, the three comparisons have these weights:

Table 7

Three contrast meanings in the background-sound study.

Comparison	Weights	Estimate
Instrumental music minus Silence	$(-1, 1, 0)$	-2
Music with lyrics minus Silence	$(-1, 0, 1)$	-4
Equal average of the two music conditions minus Silence	$(-1, 1/2, 1/2)$	-3

Only the first two form the declared family whose intervals Section 3.3 calibrates together. The third comparison gives the two music conditions equal weight. It is not a substitute for both separate comparisons when both are scientifically required. In unequal groups, equal method weights and weights proportional to student counts answer different questions. The present groups have equal counts, so the two averaging rules coincide; Appendix B develops the distinction.

A grand mean with weights $(1/3, 1/3, 1/3)$ is a useful linear combination but not a contrast, because those weights sum to 1. Multiplying every contrast weight by two doubles the estimate and SE. It preserves a zero-null t statistic, but changes the quantity being reported. Naming the comparison in ordinary words prevents a correct calculation from answering the wrong question.

3.2 Calculate the uncertainty for that comparison

The SE of a contrast extends the two-group SE of Chapter 3, Section 2.2. A difference between two means is the contrast with weights $+1$ and -1 , and Chapter 3's pooled SE can be written with those weights visible:

$$s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_0}} = \sqrt{s_p^2 \left(\frac{1^2}{n_1} + \frac{(-1)^2}{n_0} \right)}.$$

Under the independent common-variance model, each estimated group mean has variance σ^2/n_j , and the variance of a weighted sum of independent means uses the **squared weights**. Opposite signs define a difference; they do not make independent sources of uncertainty cancel. Several groups make two changes to Chapter 3's form: the weights c_j can describe any comparison, and s_p^2 is replaced by MS_W , the variance pooled over all J groups (Section 2.4). Then

$$\widehat{SE}(\widehat{C}) = \sqrt{MS_W \sum_{j=1}^J \frac{c_j^2}{n_j}}.$$

For each difference with Silence,

$$\widehat{SE} = \sqrt{\frac{178}{27} \left(\frac{1}{10} + \frac{1}{10} \right)} = 1.1483 \text{ points.}$$

Each contrast uses the three-group estimate 6.5926 and its 27 error df, although it uses only two of the three means: under the common-variance model all three groups estimate the same σ^2 , so pooling them uses more information than the two-group refit of Section 2.4, whose 7.0000 on 18 df gives an SE of 1.1832. For the equal-average contrast, the squared weights 1, 1/4 and 1/4 give $\widehat{SE} = 0.9944$.

This smaller SE belongs to a different estimand, an average of two means compared with the reference. It does not establish that either separate comparison is more precise than before. The NIST contrast explanation develops the same weighted-mean variance calculation.

3.3 Families of comparisons and the familywise error rate

A study seldom asks only one particular question, and a reader needs to know how many comparisons were made and when each was chosen. A **planned comparison** is declared before the results are seen, as this study's plan declares each music condition against Silence. A **post hoc comparison** is chosen after the results are seen, often because it looks large, such as the largest of the three pairwise differences. Chapter 3, Section 7.3, showed that a result chosen because it looks large tends to overstate its target; the same selection problem arises among the several comparisons of one study.

Each comparison tested at level .05 has a **per-comparison error rate**, also called the testwise rate, of .05: when its null hypothesis is true, it rejects wrongly with probability .05. A reader of the whole report, however, cares whether any of its comparisons is a false rejection. The **familywise error rate** is the probability of at least one Type I error (Chapter 3, Section 7.2) among all the comparisons of a family. For m independent comparisons whose null hypotheses are all true, each tested at .05, it is $1 - (1 - .05)^m$, which Figure 3 draws against m : already .1426 for three comparisons and .2262 for five.

To hold the familywise rate at .05, **Bonferroni's method** tests each of the m comparisons at level $.05/m$, or builds each interval with confidence $1 - .05/m$. Its guarantee needs no independence, because the probability that at least one of several events occurs is at most the sum of their probabilities:

$$P(\text{at least one false rejection}) \leq \sum_{k=1}^m P(\text{false rejection } k) = m \cdot \frac{.05}{m} = .05.$$

The lower line of Figure 3 stays at or below .05 for every m . The price is width: each interval is wider and each comparison has less power, so a family should contain the comparisons that matter rather than every comparison that could be made. Bonferroni's method suits any declared set; other methods are calibrated for particular families and give narrower intervals when their family fits the question (Table 8) (Kutner et al., 2005; Tukey, 1953).

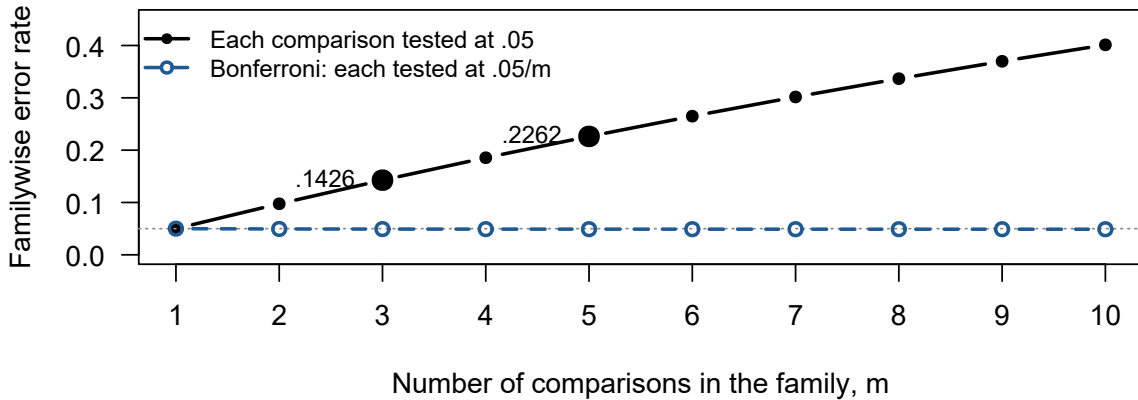


Figure 3: The familywise error rate for m hypothetical independent comparisons whose null hypotheses are all true, each tested at .05 (upper line) or at Bonferroni's .05/ m (lower line).

Table 8

Families of comparisons and the methods that hold the familywise error rate for each.

Family of comparisons	When it fits	Method that holds the familywise rate
One comparison	The plan names it in advance as the study's only question	None needed: the ordinary 95% interval, $\widehat{C} \pm t_{.975} \widehat{SE}$
A declared set of m comparisons	The plan names the set before the data are seen	Bonferroni: each comparison at .05/ m
Each group against one reference	The plan compares every condition with a control or reference condition	Dunnett's method
All pairs of groups	Every pairwise difference matters, or a pair is picked after the means are seen	Tukey's method
Every contrast, including averages of groups	Any comparison may be formed after the data are seen	Scheffé's method

The wider the family, the more comparisons its intervals must cover together, and the larger the critical value. For one difference between two sound conditions, on 27 error df, the value for a single planned comparison is 2.0518 and Scheffé’s value for every contrast of the three means is 2.5900; the declared pair, Tukey’s all pairs and Bonferroni’s three pairs fall in between. The family comes from the comparisons for which coverage is owed, fixed before looking at which interval excludes zero. An omnibus rejection is not required before a planned comparison, and it does not by itself protect comparisons chosen afterward.

The study plan declares two comparisons, each music condition against Silence, so the family is that pair, and Bonferroni gives each interval 97.5% confidence, with .0125 in each tail: the critical value is $t_{.9875;27} = 2.3734$. Table 9 gives the two intervals, $\widehat{C} \pm 2.3734 \widehat{SE}$.

Table 9

Bonferroni 95% simultaneous intervals for the two declared comparisons with Silence.

Comparison	Estimate	SE	Family lower	Family upper
Instrumental music minus Silence	−2	1.1483	−4.7253	0.7253
Music with lyrics minus Silence	−4	1.1483	−6.7253	−1.2747

The two estimates share the Silence mean, so they are positively related: a Silence mean that happens to fall high lowers both differences together, and their covariance under independent group means is $\text{Var}(\bar{Y}_S) = \sigma^2/10$. The union bound needs no independence, so the joint coverage of at least 95% stands; it describes repeated use of the whole procedure, not a probability assigned to the two realized intervals. Dunnett’s method, calibrated for exactly this family of comparisons with one reference, would give slightly narrower intervals; Bonferroni’s arithmetic is the more transparent. The NIST Bonferroni reference explains the joint-coverage bound.

These intervals read with the one-way box of Section 2.3, with three changes. The target becomes two contrasts in comprehension points, $\mu_I - \mu_S$ and $\mu_L - \mu_S$, rather than the equality of all three means; the choices add the declared family and Bonferroni’s calibration, fixed before the scores were seen, which only the analysis record can confirm; and the bound belongs to this declared pair, not to comparisons chosen after the means were inspected.

The omnibus F and these intervals answer different questions. The F test says that the three model means are not all equal; the intervals say which declared differences are established and how large they may be. Jointly, Music with lyrics minus Silence is negative, between −6.7253 and −1.2747 points. The Instrumental-music interval, $[-4.7253, 0.7253]$, includes zero: it is compatible with a small benefit, no difference or a loss of several points, so it establishes no equivalence, which would need a declared margin and an analysis designed for it. These are calculations for invented scores, not a finding about studying with music.

4 Multi-Way (Factorial) ANOVA

Analysis of variance with two or more crossed factors is **multi-way**, or **factorial**, ANOVA; this section works the two-way case in full with the feedback-by-practice study and closes, in Section 4.7, with three and four factors.

4.1 What interaction means, and why study two factors at once

Two factors **interact** when the effect of one factor differs across the levels of the other. Nothing more technical is required to state the idea. A combination of levels then produces a result that the two separate comparisons would not have led us to expect.

Here is one invented illustration, outside the chapter's two studies. Suppose a short vocabulary preview is given before a reading passage, and that students are also grouped by whether they began with weak or strong vocabulary. The preview raises comprehension scores substantially for the students who began weak, and barely changes them for the students who began strong. The effect of the preview *depends on* prior vocabulary: that dependence is the interaction. Describing only an average preview effect would tell a reader that the preview helps somewhat, and would conceal that it helps one group a great deal and the other hardly at all.

Studying two factors together in one crossed design has three advantages over running two separate one-factor studies.

1. **It is more efficient.** Every student contributes to both factor comparisons at once, because the levels of one factor are spread across the levels of the other. A crossed design therefore needs fewer cases per condition than a series of separate one-factor studies aimed at the same comparisons.
2. **It reduces residual variation.** A second factor that influences the outcome moves variation out of the error term and into the fit. If all 24 students had been analyzed with feedback as the only factor, the practice and interaction components, 96 and 64, would have stayed inside the residual term instead of leaving the 60 within-cell error used below. Section 4.4 shows what leaving practice out does to these scores.
3. **It makes interaction examinable.** Only a crossed design supports the question of whether a comparison changes across the other factor's levels. A series of one-factor studies supplies no estimate of that change, however carefully each one is run.

In the feedback-by-practice study introduced in Section 1, practice schedule therefore gives each feedback comparison a context. We can compare feedback conditions within Massed practice, within Spaced practice, or after averaging over the two schedules. These are distinct quantities. A **main effect** compares one factor's levels after averaging across the other factor with stated weights. A **simple effect** makes the comparison at a specified level of the other factor. **Interaction** means that a comparison changes across those levels.

4.2 Main effects and interaction through graphs

In a two-factor design either main effect, the interaction, or any combination of the three can be present. Six schematic two-by-two patterns make the whole range visible before we return to the full three-by-two data. Each line in Figure 4 represents a feedback level, A or B; the horizontal positions represent Massed and Spaced practice. These are mean-only drawings, not additional observed studies, and they carry no significance claim.

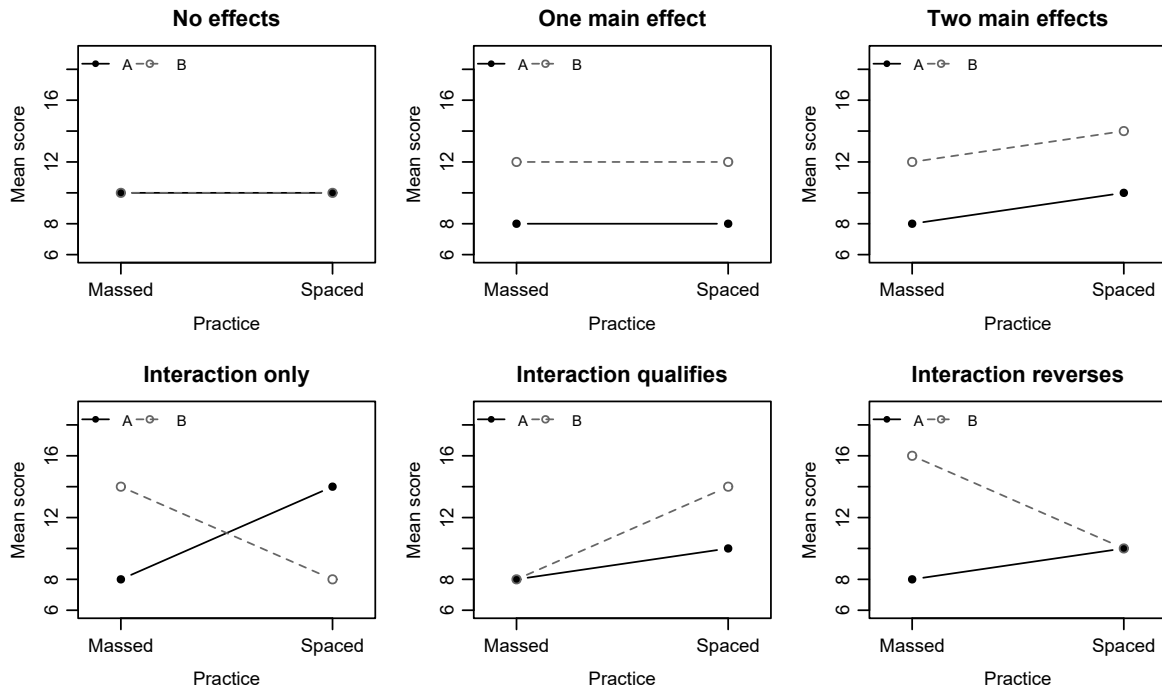


Figure 4: Six schematic patterns of cell means, covering the whole range of main-effect and interaction combinations. Lines identify feedback levels A and B; the drawings illustrate averages and changing differences, without significance claims.

Read each pattern by taking differences, not only by recognizing a shape. Define a schedule difference as Spaced minus Massed and a difference of differences as B's schedule difference minus A's.

Table 10*Arithmetic behind the six schematic patterns.*

Pattern	A: Massed, Spaced	B: Massed, Spaced	Schedule differences A, B	Difference of differences
No effects	10, 10	10, 10	0, 0	0
One main effect	8, 8	12, 12	0, 0	0
Two main effects	8, 10	12, 14	2, 2	0
Interaction only	8, 14	14, 8	6, -6	-12
Interaction qualifies	8, 10	8, 14	2, 6	4
Interaction reverses	8, 10	16, 10	2, -6	-8

In the first panel nothing varies: both feedback levels and both schedules have mean 10, and the two lines coincide. In the second, B exceeds A by 4 points under either schedule while the schedule comparison is 0 under either feedback level; there is one main effect and no interaction. In the third, B still exceeds A by 4 points everywhere, and Spaced now exceeds Massed by 2 points everywhere: two main effects, again with a zero difference of differences. Parallel lines express those constant comparisons.

The last three panels have a nonzero difference of differences. In the crossing panel, both feedback marginal means are 11 and both practice marginal means are 11 when cells receive equal weight. Both averaged comparisons are zero despite large conditional differences, which is why a small main effect need not mean that a factor is irrelevant at every level of another factor.

In the qualifying panel the schedule comparison is 2 for A and 6 for B. Both are positive, so the averaged 4-point schedule comparison points the right way for both feedback levels but describes neither exactly. The lines need not cross for interaction to be present. In the reversing panel the schedule comparison is 2 for A and -6 for B, so the averaged comparison of -2 points the wrong way for A. Reporting only the two main effects would then mislead a reader about every conditional comparison. Examine the interaction first, and read a main effect as an average whose parts are still visible.

These drawings establish only the numerical patterns: significance also needs information about variation, sample size and the reference distribution.

4.3 Cells, marginal means and the full data

Turn to all 24 students of the feedback-by-practice study. Each of the six cells contains four different students; no person contributes a score under the other practice schedule. This analysis therefore uses six separate groups of four, where the background-sound analysis used three groups of ten.

Write $\bar{y}_{jk}$ for a cell mean, with $j = 1, 2, 3$ indexing feedback and $k = 1, 2$ indexing practice. A dot replaces an index that has been averaged over: $\bar{y}_{.j}$ averages a feedback level over practice, and $\bar{y}_{.k}$ averages a practice level over feedback. An **effect** in this layout is a marginal mean's departure from the grand mean, 10. Table 11 gives the cell means, the equally weighted marginal means and the effects.

Table 11

Cell means, equally weighted marginal means and effects for the complete invented study.

Feedback	Massed	Spaced	Feedback mean	Feedback effect
Rereading	6	6	6	−4
Answer feedback	8	12	10	0
Explanatory feedback	10	18	14	4
Practice mean	8	12	Grand mean 10	
Practice effect	−2	2		

Each marginal mean averages its cell means with equal weights; with four students in every cell it is also the mean of that condition's students. The feedback effects, −4, 0 and 4, and the practice effects, −2 and 2, each sum to zero. The practice comparison, Spaced minus Massed, averages 4 points over the three feedback conditions, but its components, the schedule differences within the rows, are 0, 4 and 8. A four-point main effect is therefore a meaningful stated average that describes none of the conditional comparisons except Answer feedback's.

The connected lines in Figure 5 organize cell means. They do not show individual students moving from one condition to another. The nonparallel pattern describes interaction in the observed means; the full model supplies the uncertainty calculation for an inferential claim.

4.4 Additivity and the full model

If feedback and practice combined additively, every cell mean would be the grand mean plus its feedback effect plus its practice effect, and the schedule difference would be the same in every row. Table 12 sets these **additive predictions** beside the observed cell means. The difference between an observed cell mean and its additive prediction is an **interaction departure**.

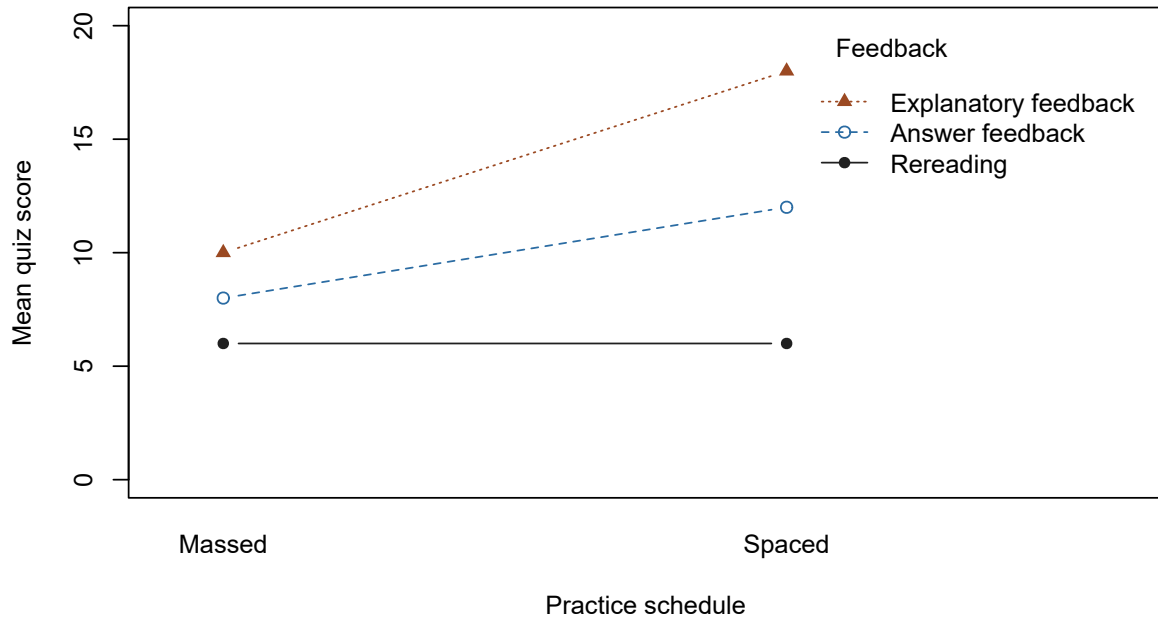


Figure 5: Quiz cell means across practice schedules in the full 3×2 example. Each line joins means of different students; the schedule differences are 0, 4 and 8 points.

Table 12

Observed cell means, additive predictions from the grand mean and both effects, and interaction departures.

Cell	Observed mean	Additive prediction	Interaction departure
Rereading, Massed	6	4	2
Rereading, Spaced	6	8	-2
Answer feedback, Massed	8	8	0
Answer feedback, Spaced	12	12	0
Explanatory feedback, Massed	10	12	-2
Explanatory feedback, Spaced	18	16	2

Answer feedback's two cells sit exactly on their predictions, but the other four do not: Rereading gains less from spacing than the average, and Explanatory feedback more, by 2 points in each cell. Agreement in one row does not make the whole layout additive.

The full mean model includes such departures:

$$Y_{ijk} = \mu + \alpha_j + \beta_k + (\alpha\beta)_{jk} + \varepsilon_{ijk}.$$

Here $i = 1, \dots, 4$ identifies a student within a cell, $j = 1, 2, 3$ identifies feedback and $k = 1, 2$ identifies practice. The grand model mean is μ ; α_j and β_k are the factor effects; $(\alpha\beta)_{jk}$ is the interaction departure; and ε_{ijk} is the individual error around the cell mean. These indexed effects are distinct from the unindexed error probabilities α and β used in hypothesis testing.

In the balanced decomposition, main effects sum to zero, and interaction departures sum to zero along each row and each column, so they carry only the pattern left after the two main effects. With two columns, a row's second departure is the negative of its first, and the three first-column departures, 2, 0 and -2 , sum to zero, so only two of them are free. Thus the interaction has $(3 - 1)(2 - 1) = 2$ df. It describes a two-dimensional pattern, rather than one particular four-cell difference.

Each score's distance from the grand mean now splits into four parts, a feedback effect, a practice effect, an interaction departure and the student's residual. Squaring the parts and adding over all 24 students gives, in this balanced layout as in Section 2.2, four sums of squares with matching df:

$$SS_T = SS_A + SS_B + SS_{AB} + SS_E, \quad df_T = df_A + df_B + df_{AB} + df_E.$$

In this two-way table A denotes feedback and B practice, so SS_B now means the practice component, not the between-group component of the one-way table, and SS_E is the within-cell error:

$$\begin{aligned} SS_A &= 2n \sum_j (\bar{y}_{j\cdot} - \bar{y})^2 = 256, \\ SS_B &= 3n \sum_k (\bar{y}_{\cdot k} - \bar{y})^2 = 96, \\ SS_{AB} &= n \sum_j \sum_k (\bar{y}_{jk} - \bar{y}_{j\cdot} - \bar{y}_{\cdot k} + \bar{y})^2 = 64, \\ SS_E &= \sum_j \sum_k \sum_i (y_{ijk} - \bar{y}_{jk})^2 = 60. \end{aligned}$$

Each component counts its effects or departures once for every student who shares them. The partition reads $476 = 256 + 96 + 64 + 60$ on $23 = 2 + 1 + 2 + 18$ df: each of the six cells uses one mean, so the error has $24 - 6 = 18$ df, and the fit part is $256 + 96 + 64 = 416$ on five df.

Table 13*Full two-way ANOVA using all 24 students.*

Source	SS	df	MS	F	p
Feedback (A)	256	2	128.0000	38.4	<.0001
Practice (B)	96	1	96.0000	28.8	<.0001
A $\times$ B	64	2	32.0000	9.6	.0015
Error	60	18	3.3333		
Total	476	23			

Read this table as in Section 2.3: the three source rows are the fit, the error row the residual and the total the data, and the sums of squares and df add across the rows. Read the interaction row first, because it decides how the two main-effect rows can be read.

The error row of Table 13 holds only the variation of students around their own cell means. Figure 6 shows what happens to that row if practice is left out of the model, so that each feedback condition becomes one group of eight students. This is not an analysis the design calls for; it shows what the second factor removes from the error term.

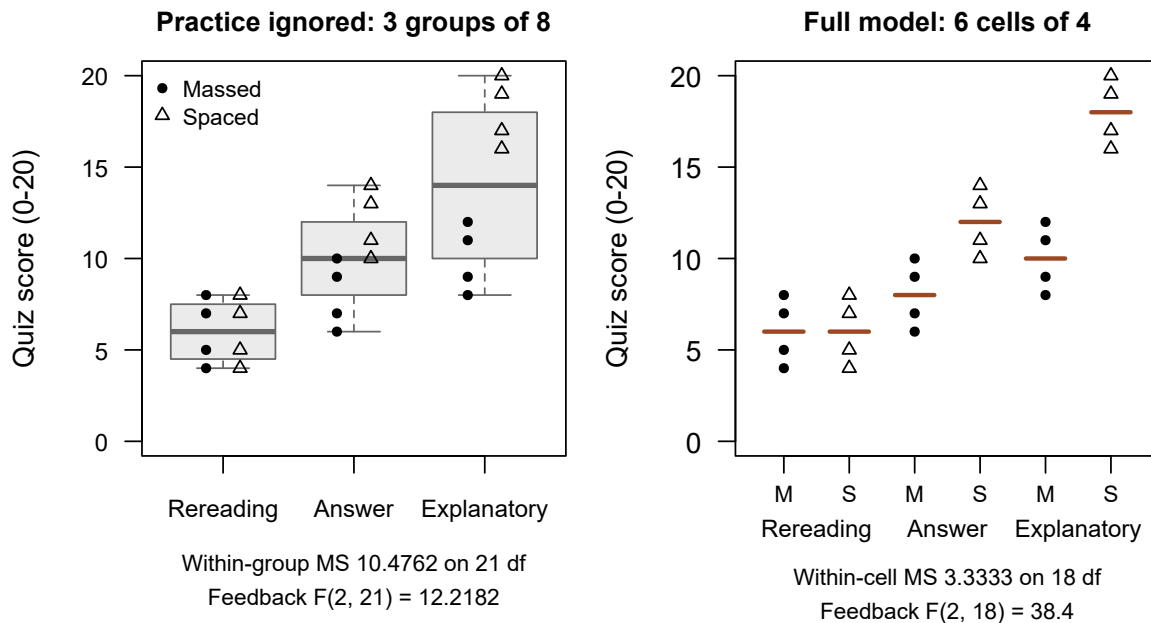


Figure 6: The 24 feedback-by-practice scores analysed two ways. Left: practice is left out, so each feedback condition is one group of eight; filled circles mark Massed and triangles Spaced students. Right: the six cells of the full model, with a bar at each cell mean. Leaving practice out moves the practice and interaction variation into the within-group term.

Table 14*What leaving practice out does to the error term.*

Analysis of the 24 scores	The within-group term holds	Error SS	Error df	Error MS	Feedback F	p
Feedback only, practice left out	Within-cell error 60, practice 96 and interaction 64	220	21	10.4762	12.2182	.0003
Full two-way model	Within-cell error 60 only	60	18	3.3333	38.4000	<.0001

Leaving practice out changes neither the feedback means, 6, 10 and 14, nor the feedback sum of squares, 256; it changes the error term. The practice and interaction components stay inside the within-group variation, which becomes $220 = 60 + 96 + 64$ on 21 df, so the within-group mean square, 10.4762, is more than three times the full model's 3.3333 and the feedback F falls from 38.4000 to 12.2182. The left panel shows where the extra variation comes from: each group mixes two schedules, so the Explanatory group splits into two clusters eight points apart and its SD, 4.5981, is 2.7203 times Rereading's 1.6903, whose two schedules do not differ. One pooled variance would misdescribe all three groups. A factor that influences the outcome belongs in the model, which is the second advantage of a crossed design in Section 4.1.

Every displayed F divides its source mean square by the same $MS_E = 60/18 = 10/3$. The corresponding null hypotheses address different features of the model:

Table 15*Three null hypotheses with their averaging rules.*

Source	Null hypothesis	Comparison represented
Feedback	$\mu_{R.} = \mu_{A.} = \mu_{E.}$	Equal feedback means, each averaging the two schedules equally
Practice	$\mu_{.M} = \mu_{.S}$	Equal schedule means, each averaging the three feedback levels equally
Interaction	$(\alpha\beta)_{jk} = 0$ for every cell	Each feedback comparison is constant across schedules, equivalently each schedule difference is the same across feedback levels

Independent common-variance Normal errors supply the classical F references. Under these conditions, the interaction result $F(2, 18) = 9.6$, $p = .0015$ gives evidence against an additive pattern. Both main effects also have small p-values, but their meanings remain averaged comparisons. Reading them as “each factor works equally well everywhere” would discard the interaction.

The clean component decomposition uses the balanced layout. With unequal cell counts, the averaging target and the model comparison need more care; different software sums of squares can address different hypotheses. The weighting reference explains the first distinction without changing these data.

💡 Justifying the analysis

- **Target:** the six cell means μ_{jk} of the feedback-by-practice layout for the 24 students, in quiz points. The three F tests ask whether the equally weighted feedback means are equal, whether the two equally weighted practice means are equal, and whether every interaction departure is zero.
- **Conditions:**
 - *Secured by design:* students individually and randomly assigned to the six cells, four different students in each, each scored once; no student contributes a score under both schedules.
 - *Model assumptions:* Normal errors with one common variance in every cell, estimated by $MS_E = 3.3333$ on 18 df. The model includes the interaction, so it does not assume additivity. A main effect remains a defined average over the other factor's levels; reading it as the effect at every level would assume additivity.
 - *Choices and defaults:* the full model with interaction; equal cell weights for the marginal means; a .05 level fixed in advance. With four students in every cell, every sums-of-squares convention gives the same table.
- **Check:** every cell has residuals $-2, -1, 1$ and 2 , so the six spreads agree by construction; residuals fanning out as the fitted mean grows would question the common variance. Four scores per cell say little about shape. The interaction row, $F(2, 18) = 9.6, p = .0015$, together with the cell-mean plot, shows that additivity fails for these scores.
- **Bound:** under the model, the evidence is against an additive pattern. The main effects, $F(2, 18) = 38.4$ and $F(1, 18) = 28.8$, both $p < .0001$, describe equally weighted averages; the schedule differences within the three feedback conditions, 0, 4 and 8 points, show what those averages conceal. The scores are invented, so the result describes a calculation, not an educational finding.

4.5 Simple effects and a selected interaction contrast

The Answer-feedback schedule comparison, Spaced minus Massed, is a **simple effect**: $12 - 8 = 4$ points. Under the full model its SE is $\sqrt{(10/3)(1/4 + 1/4)} = 1.291$, and with 18 error df a pointwise 95% interval is $[1.2877, 6.7123]$. This interval covers the one stated simple comparison; it is not protected for every simple effect that a reader might inspect after seeing the graph.

The two means come from different students, so this is an independent-cell calculation, not a paired one.

Now suppose a reader selects the Answer-versus-Rereading change in the schedule comparison after inspecting the pattern. In model means the target is

$$C = (\mu_{A,S} - \mu_{A,M}) - (\mu_{R,S} - \mu_{R,M}),$$

one **difference of differences**, estimated as $(12 - 8) - (6 - 6) = 4$. Its four cell means come from different students and carry weights 1, -1 , -1 and 1, so squaring the weights gives $\widehat{SE}(\widehat{C}) = \sqrt{(10/3)(1/4 + 1/4 + 1/4 + 1/4)} = 1.8257$. Because the contrast was chosen after the graph was seen, its interval must cover every interaction contrast that could have been chosen: Scheffé's family for **all contrasts in the two-dimensional interaction space**, with multiplier $\sqrt{2F_{.95;2,18}} = 2.6663$, gives $4 \pm 2.6663(1.8257) \approx [-0.868, 8.868]$.

This interval reads with the contrast sentences of Section 3.3, with these changes: the design and check are those of the factorial box in Section 4.4; the target is a difference of differences among four cell means; the error estimate is the full model's $MS_E = 3.3333$ on 18 df; and the family is Scheffé's, because the comparison was selected after inspection.

The interval includes zero even though the omnibus interaction p-value is .0015. There is no contradiction. The joint test assesses the entire two-dimensional departure from additivity; the selected contrast concerns one direction, with protection for choosing among all interaction contrasts. Evidence that some interaction direction is nonzero does not establish every selected direction. Had the plan named this one contrast before the scores were seen, the pointwise interval $[0.1643, 7.8357]$ would be warranted, and it excludes zero; choosing it now because it excludes zero would discard the protection that the selection requires. Here the history of the comparison decides the conclusion.

The protected space is specific. The simple Answer schedule effect contains a practice main-effect component as well as an interaction component, so it lies outside this interaction-only family, and all contrasts among the six cell means would form a larger, five-dimensional space with a larger multiplier. A claim covering those comparisons would need its own family.

4.6 Give a connected interpretation

The observed schedule differences are 0, 4 and 8 quiz points for Rereading, Answer feedback and Explanatory feedback. The full-model interaction test gives $F(2, 18) = 9.6$, $p = .0015$ under the stated conditions. The selected Answer-versus-Rereading change in schedule difference is 4 points, with a 95% Scheffé interval $[-0.868, 8.868]$ over all interaction contrasts. The overall result does not establish that particular selected comparison.

Feedback and practice marginal means summarize explicit equal weights. The cell means show what those averages can conceal. The one-way conclusions of Sections 2 and 3 concern

the background-sound study, whereas this factorial conclusion uses all 24 students of the feedback-by-practice study and 18 error df. These are analyses of invented data, not empirical evidence that a teaching method succeeds in a population of students.

4.7 Three and four factors

Suppose a larger version of the feedback-by-practice study added a third factor, quiz timing: students take the quiz either right after their final practice or one week later. Crossing timing with the six feedback-by-practice cells gives twelve cells, a three-way layout, with four different students in each cell. The extension is hypothetical, and Figure 7 draws cell means only: it shows a pattern, not data, and carries no significance claim.

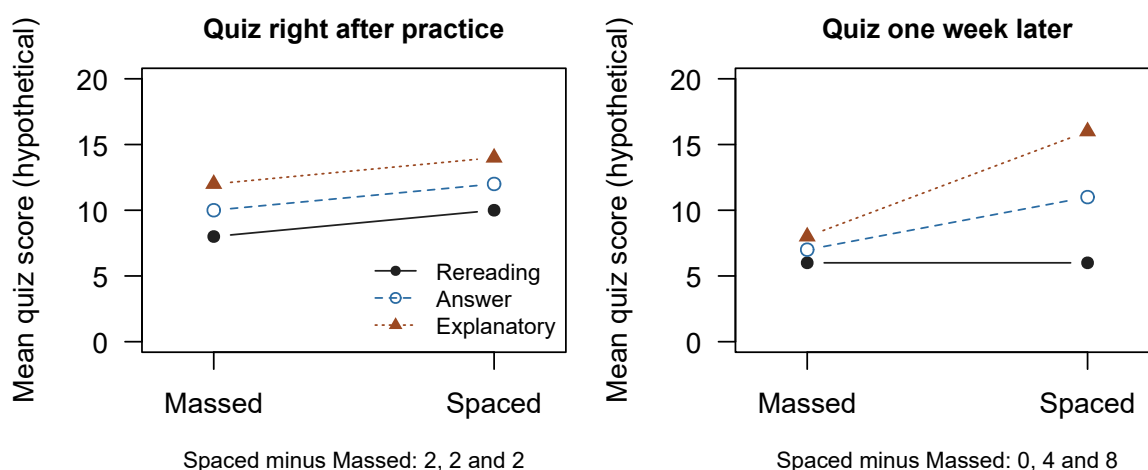


Figure 7: A hypothetical three-way pattern, drawn as cell means only. Each panel is the feedback-by-practice plot for one quiz timing. Right after practice the lines are parallel, so Answer’s schedule difference minus Rereading’s is 0; one week later they fan out, and the same difference of differences is 4. That change across timing is a three-way interaction.

Read each panel as Figure 5 was read. Right after practice the three lines are parallel: spacing adds 2 points under every kind of feedback, so Answer’s schedule difference minus Rereading’s is $2 - 2 = 0$, and this panel shows no two-way interaction. One week later the lines fan out: spacing adds 0, 4 and 8 points, and the same difference of differences is $4 - 0 = 4$. The feedback-by-practice interaction therefore differs between the two timings, and that change is a **three-way interaction**: a two-way pattern that changes across the levels of a third factor. One three-way contrast is the change in the difference of differences, $4 - 0 = 4$ points, a difference of differences of differences.

A fourth factor carries the idea one level further. Suppose the study also crossed reading on paper or on screen with the twelve cells, each student reading the practice material in one medium, again with four different students in each of the 24 cells. A **four-way interaction** would mean that the three-way pattern itself differs between the two media: for example, that the pattern of Figure 7, parallel lines right after practice and lines fanning out by feedback a week later, holds for students who read on paper, while for students who read on screen the lines stay parallel at both timings. Table 16 counts what each added factor brings when every cell keeps four different students.

Each added two-level factor doubles the cells, so four students per cell means 24, then 48, then 96 students. The effects grow faster than the factors: a design with k factors has k main effects and one interaction for every pair, triple and larger set of factors, $2^k - 1$ sources in all, so a three-factor table has seven rows of effects and a four-factor table fifteen.

Read such a table from the highest-order interaction down. Section 4.4 read the interaction row before the main effects, because a main effect averages over a pattern that the interaction says is changing. One level up the rule is the same: when a three-way interaction is present, each two-way interaction is an average over the levels of the third factor. Averaged over the two timings of Figure 7, Answer's schedule difference minus Rereading's is 2 points, which describes neither panel. Report the two-way patterns within each level of the third factor instead, and read a lower-order row alone only when the design and the evidence make the higher-order interactions negligible (Kutner et al., 2005).

Table 16

Cells, students, effects and error df as quiz timing and then reading medium are added to the feedback-by-practice design, four students per cell.

Design	3×2	$3 \times 2 \times 2$	$3 \times 2 \times 2 \times 2$
Cells	6	12	24
Students, four per cell	24	48	96
Main effects, one per factor, k	2	3	4
Two-way interactions	1	3	6
Three-way interactions	0	1	4
Four-way interactions	0	0	1
Sources of effects, $2^k - 1$	3	7	15
Error df, students minus cells	18	36	72

Added factors have costs. A three-way contrast combines eight cell means where a difference of differences combines four, so with the same students per cell and the same error variance its SE is $\sqrt{2}$, about 1.4, times as large, and matching the precision of a difference of differences takes twice as many students per cell. The seven F tests of a three-factor table, each at .05, also form a family in the sense of Section 3.3: for seven independent tests whose null hypotheses are all true, the chance of at least one false rejection is .3017, where the upper line of Figure 3

stands at $m = 7$, and the fifteen tests of a four-factor table raise it further. More cells also make unequal or empty cells more likely: unequal counts make each marginal mean depend on its weights (Appendix B.1), and an empty cell leaves some interactions without an estimate.

A study with several factors therefore declares in advance the few comparisons it was built for, such as whether the spacing benefit under Answer feedback grows between the two timings, and reads the other rows and cells as description. Twelve cells allow 66 pairwise comparisons and many more contrasts, so a comparison found by searching them needs a family that covers the search, as Section 3.3 showed for three groups.

5 Checking the Conditions

5.1 What the design and the data can show

Sections 2–4 stated each condition where a procedure first relied on it, and each *Justifying the analysis* box recorded what the data could check. Table 17 gathers the conditions of the chapter’s analyses, what the design or the data can show about each, and the response when one fails.

A sequence of Normality or variance tests should not become an automatic permission rule: such tests have least power in the small samples where the choice matters most. A condition is judged from the design, subject knowledge and displays such as Figure 8 together (Kutner et al., 2005).

Table 17

Conditions of the chapter's analyses, what can show whether each holds, and the response if it fails.

Condition	What the design or the data can show	Response if it fails
Independent observations	Only the design record: individual assignment, individual sessions, one score per student. No display of the scores can establish it	An analysis that represents the dependence: comparisons within blocks, as in Chapter 3's paired schools (Section 2.3), or models for classes and repeated measurements. Treating dependent rows as independent misstates every SE; positive clustering makes it too small, by about the square root of the design effect of Chapter 2, Section 4.4
Roughly Normal errors around the group or cell means	A residual display and a Normal quantile plot of the residuals, not of the pooled scores; small groups reveal only pronounced departures	With moderate skew and larger groups the F reference remains a reasonable approximation; with small groups and strong skew or a dominating value, disclose the weaker approximation and examine the influential scores
One common variance	The group or cell SDs beside the group sizes; a fan in the residuals as the fitted mean grows	Welch's one-way ANOVA, chosen before the analysis (Section 5.2); for a contrast, an SE built from its groups' own variances, which for a pair is Chapter 3's Welch t
Additivity, when a main effect is read as the effect at every level	The interaction row and the cell-mean plot	Report cell means and simple effects, and read each main effect as a stated average (Section 4)

For the background-sound study the residuals spread similarly around zero in all three groups, each from -4 to 4 , and in the quantile plot they follow the line, with no outlying value; the residuals of -4 and 4 at the two ends sit a little flatter than a Normal tail. Thirty residuals can show only pronounced departures, so the display is consistent with the model rather than proof of it, and independence still rests on the design. The factorial scores were built with residuals -2 , -1 , 1 and 2 in every cell, a teaching construction that no display could treat as evidence.

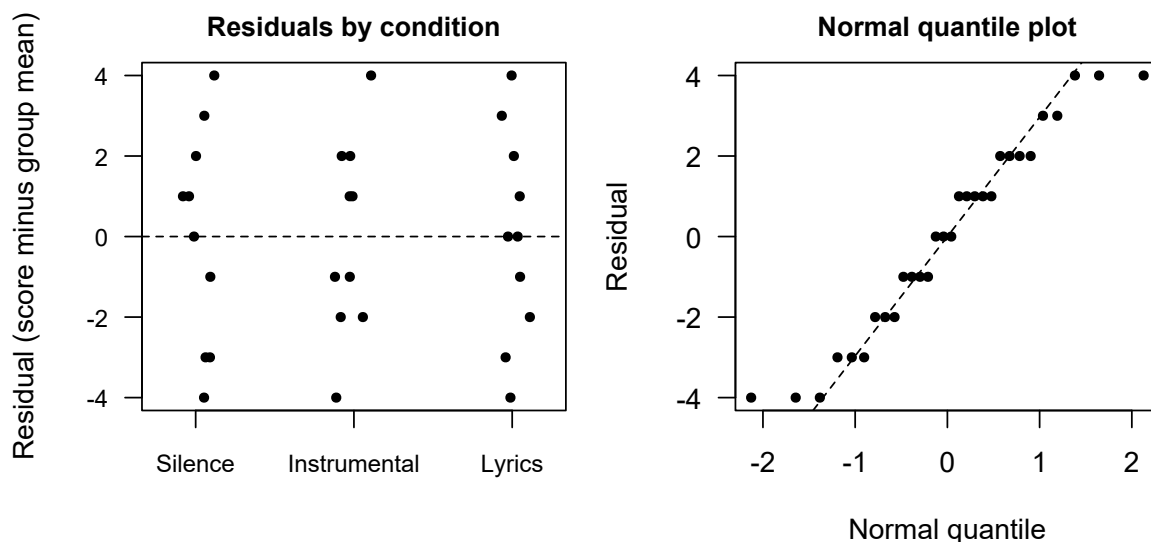


Figure 8: Residuals of the one-way fit to the background-sound scores. Left: each student’s residual, the score minus the group mean, by condition, jittered sideways so that equal values stay visible. Right: the thirty residuals against Normal quantiles, with a reference line.

Missing outcomes raise a further question that neither the model nor a display answers. A report compares assigned and analysed counts and explains known reasons for losses, because missing outcomes can change who contributes to a comparison and can require assumptions beyond the ANOVA model. Neither invented dataset has missing scores: all 30 background-sound scores and all 24 feedback-by-practice scores are observed. Keep three claims distinct in every report: the observed means and sums of squares describe these records; intervals and tests express uncertainty under a stated model; and causal or population interpretations need support from allocation, participation, measurement and the intended scope.

5.2 When spreads differ: Welch’s one-way ANOVA

When a common variance is doubtful, **Welch’s one-way ANOVA** is the counterpart of Welch’s unequal-variance t test in Chapter 3, Section 5.4. It keeps each group’s own variance instead of pooling them into MS_W , weights each group mean by the precision implied by its own variance and size, n_j/s_j^2 , and uses an approximate F reference whose denominator df are estimated from the group variances (Welch, 1951); with two groups its F is the square of Welch’s t. It reads with the one-way box of Section 2.3, with two changes: the model assumption of one common variance is dropped, so each group mean needs only a near-Normal sampling distribution, and the calibration changes with it. For the background-sound study it gives $F(2, 17.9561) = 5.5144$, $p = .0136$, beside the classical $F(2, 27) = 6.0674$, $p = .0067$: with similar spreads and equal group sizes the two calibrations lead to the same conclusion at

.05, although Welch's p is about twice as large. Its F is slightly smaller, and its denominator df fall from 27 to 17.9561 because each weight rests on a variance estimated from its own ten scores. Welch's method has the scope of one factor: clustering still needs an analysis that represents the design, and crossed factors with unequal cell spreads take each stated contrast's SE from the cells' own variances.

The classical F test applies one pooled variance to every group mean. Chapter 3, Section 5.4, showed for two groups that a pooled variance misstates the variance of a small group's mean when spreads differ. Table 18 shows what this does with three groups: each row simulates 50,000 studies of three Normal groups with equal means and the stated SDs, so every rejection at level .05 rejects a true null.

Table 18

Rejection rates of a true null of equal means at nominal level .05, from 50,000 simulated studies per design.

Group sizes	Population SDs	Classical F	Welch's one-way ANOVA
10, 20 and 40	8, 4 and 4: the smallest group varies most	.1572	.0511
10, 20 and 40	4, 4 and 8: the largest group varies most	.0126	.0499
20, 20 and 20	8, 4 and 4	.0610	.0502

When the smallest group varies most, the variance that pooling targets, 22.4478, gives that group's mean a variance of 2.2448 instead of the 6.4000 its own SD implies, and the classical F rejects the true null in .1572 of the studies, more than three times the nominal .05; when the largest group varies most it rejects in only .0126, and with equal group sizes it rises only to .0610. Welch's one-way ANOVA stays near .05 in all three designs, which is why unequal group sizes with spreads that may differ call for it.

As in Chapter 3, choose between the common-variance and Welch calibrations before the analysis, from the design and subject knowledge, rather than from a preliminary variance test or from whichever p -value is smaller (Delacre et al., 2017; Delacre et al., 2019). Many statistical programs report Welch's t by default for two groups; that default is Chapter 3's last t procedure, not the equal-variance test whose square equals the classical F of Section 2.4. Chapter 6 runs the Welch alternative beside the common-variance calculation.

6 Reading Bridge

Comparing several means extends the estimate-and-SE reasoning of Chapter 3. ANOVA separates sources of variation to ask an omnibus question; a contrast states a narrower question and its uncertainty. Main effects average across factor levels, while cell means and simple effects show the conditional pattern, and a third factor can change a two-way pattern itself. A joint interaction test and one selected interaction contrast can therefore lead to different, compatible conclusions.

The same group-mean model can be written using indicator variables, as Chapter 6 does in R. A measured baseline can also enter an adjusted model, called analysis of covariance (ANCOVA), which Chapter 9, Section 2, develops as regression after Chapters 7–8 develop the regression mathematics, several controls and standardized predictions. Baseline adjustment accounts for a measured source of score variation and compares fitted outcomes at a stated common baseline value, or averages them over the same stated baseline distribution. The groups need adequate overlap in those baseline values, often called common support, for those comparisons to be informed by observations. Adjustment is not a general cure for confounding or missing outcomes.

Blocking organizes comparisons within allocation groups, as in Chapter 3’s paired schools (Section 2.3). Repeated measurements and classroom clustering change the covariance structure and the relevant error term. The shape of a data table cannot settle these design questions. Chapters 5–6 now make the statistical foundations executable while keeping the question, records and interpretation beside each calculation.

Key terms

Use this reference to revisit the concepts principally explained here. The Part One Concept Reference collects earlier terms, including estimand, SE and confidence interval.

Table 19

Selected concepts for group comparisons and ANOVA.

Term and common alternatives	Meaning
Factor (grouping factor)	In ANOVA, a categorical variable defining groups or conditions; its categories are levels. This meaning differs from a latent factor used to model shared indicator variation.
Analysis of variance (ANOVA)	A method for studying mean differences through sources of variation. One-way ANOVA uses one grouping factor, and multi-way (factorial) ANOVA uses two or more crossed factors, as two-way and three-way ANOVA do; each factor can have several levels.
Fitted value	The value a fitted model assigns to an observed case. In a one-way group-mean model, it is that case’s group mean.

Term and common alternatives	Meaning
Residual	An observed outcome minus its fitted value. A residual is calculated from the fitted model; an error is the corresponding unknown departure in the population model.
Mean square (MS)	A sum of squares divided by its degrees of freedom. In classical ANOVA, the within-group mean square estimates the common error variance under the model's conditions.
F statistic (F ratio)	In classical ANOVA, a ratio of the mean square for a tested source to the appropriate error mean square. Its null F reference depends on the model and both degrees of freedom.
Omnibus test	A test of one joint null hypothesis about a whole pattern, such as equality of all group means. Rejecting it does not identify which comparisons differ or how large they are.
Eta squared	The proportion of total observed variation attributed to a specified source in an ANOVA decomposition. In one-way ANOVA, it is between-group divided by total sum of squares.
Contrast (linear contrast)	A weighted combination of means whose weights sum to zero, expressing a specific comparison. The means, weights, and target population determine the quantity compared.
Familywise error rate (FWER)	The probability of at least one Type I error among a defined family of tests. Procedures such as Bonferroni allocate the error allowance across that family.
Simultaneous confidence intervals	A set of intervals calibrated so that, under their conditions, all of them cover their estimands together at the stated familywise rate; Bonferroni, Tukey, and Dunnett are different calibrations for different families.
Cell mean and marginal mean	In a two-factor layout, a cell mean is the mean of the cases sharing one level of each factor; a marginal mean averages the cell means across the levels of the other factor, with stated weights.
Main effect	A comparison across one factor's levels averaged over the other factor's levels, using stated weights. It remains an average when the conditional pattern differs across levels.
Interaction	A pattern in which a comparison across one factor's levels changes with the other factor's level. A difference of differences expresses one such comparison.
Simple effect	A comparison across one factor's levels at a specified level of the other factor. It describes a conditional comparison within the factorial design.
Three-way interaction (higher-order interaction)	A pattern in which a two-way interaction changes across the levels of a third factor. Each added factor multiplies the cells and adds interactions, so the highest-order interaction is read first and a few comparisons are declared in advance.

Further reading

- *Introduction to Modern Statistics* develops the transition from grouped observations to inference for many means (Çetinkaya-Rundel & Hardin, 2024).
- Penn State STAT 200, Lesson 10 provides an introductory ANOVA treatment; its factorial-design lesson develops main effects and interaction.
- The NIST handbook explains contrasts and their SEs and Bonferroni simultaneous intervals.

Exercises

The supplied scores, tables and results support each interpretation, with a calculator for the few calculations that the reasoning needs. Retain full precision until reporting calculated results to four decimal places. State the relevant records, conditions and comparison direction in each interpretation.

Exercise 1: Read grouped observations (Foundational)

Use the thirty background-sound scores from the invented study:

- Silence: 8, 9, 9, 11, 12, 13, 13, 14, 15, 16.
 - Instrumental music: 6, 8, 8, 9, 9, 11, 11, 12, 12, 14.
 - Music with lyrics: 4, 5, 6, 7, 8, 8, 9, 10, 11, 12.
1. The group means are 12, 10 and 8, and the sample SDs are 2.7080, 2.4037 and 2.5820. Use them with the ranges to compare the centers, spreads and overlap of the three groups, and say what an SD of about 2.5 points describes for these students.
 2. Suppose every student's distance from their group mean were doubled, so that the three means stay at 12, 10 and 8. Before calculating, say whether the evidence against equal means would become stronger or weaker, and why. Then compare the supplied results, $F(2, 27) = 6.0674$, $p = .0067$ for the actual scores and $F(2, 27) = 1.5169$, $p = .2375$ for the doubled version, and explain why F falls to one quarter.
 3. The three SDs are similar. Explain why that similarity does not establish that a common-variance Normal model will hold in a real study.
 4. Identify the assignment and measurement units stipulated in this invented study. Explain what the scores alone cannot establish about causal interpretation or generalization.

Exercise 2: Read the one-way table (Foundational)

Use Exercise 1's scores. There are $J = 3$ groups and $N = 30$ students, and the classical analysis assumes independent Normal errors with a common variance. Statistical software reports this table:

Source	SS	df	MS	F	p
Sound condition	80	2	40.0000	6.0674	.0067
Within groups	178	27	6.5926		
Total	258	29			

1. Explain what the between-groups df, 2, and the within-groups df, 27, each count, and why they add to the total df, 29. Name the reference distribution of F under the null.
2. Explain why the Instrumental-music group contributes nothing to SS_B although its scores vary. Say what MS_W estimates and why it is smaller than the total sample variance, $258/29 = 8.8966$.
3. State the omnibus null. Interpret $p = .0067$ at $\alpha = .05$, including the model conditions and what the conclusion leaves unanswered about particular comparisons.
4. Without refitting, predict how SS_B , SS_W and F would change if (a) every Music-with-lyrics score rose by 2 points, or (b) all thirty scores rose by 5 points. Then check prediction (a) against the supplied result $SS_B = 26.6667$, $F(2, 27) = 2.0225$, $p = .1519$, and say what change (a) makes to the conclusion of part 3.

Exercise 3: Define and estimate contrasts (Foundational)

The sound-condition means are 12, 10 and 8 in the order Silence, Instrumental music, Music with lyrics. Each group has ten students and the common-variance estimate is $MS_W = 178/27$, with 27 residual df.

1. Identify which weight vectors define contrasts: $(-1, 1, 0)$, $(-1, 0, 1)$, $(-1, 1/2, 1/2)$ and $(1/3, 1/3, 1/3)$. State the ordinary-language question represented by each contrast.
2. The vectors $(-1, 1, 0)$, $(-1, 0, 1)$ and $(-1, 1/2, 1/2)$ give estimates -2 , -4 and -3 with SEs 1.1483, 1.1483 and 0.9944, each from $\sqrt{MS_W \sum_j c_j^2/n_j}$. Explain, through the squared weights, why the equal-average contrast has the smallest SE, and why all three SEs use the full three-group error estimate.
3. Explain why the equal-average contrast cannot replace both separate music-versus-Silence comparisons when both are required.
4. For a weighting illustration only, suppose the Instrumental-music and Music-with-lyrics means stay at 10 and 8 but their counts are 10 and 30. The Silence mean stays at 12. Compare the equally weighted music-average difference with the student-weighted difference. State the question answered by each. These illustrative counts do not change the chapter's actual balanced dataset.

Exercise 4: Interpret comparison families (Integrative)

In the invented background-sound study, thirty first-year students were individually and randomly assigned to three sound conditions, ten to a condition, and each answered the 0–20 comprehension quiz once after reading the same passage. The group means are 12 for Silence, 10 for Instrumental music and 8 for Music with lyrics. Under the independent common-variance Normal model, $MS_W = 178/27$ on 27 error df, so each difference between two of these means has $SE = \sqrt{(178/27)(1/10 + 1/10)} = 1.148267616$; this SE and the critical values below carry nine decimals so that the printed four-decimal limits can be reproduced by hand. Two reports give the same estimate, -4 comprehension points for Music with lyrics minus Silence, but the comparison has a different history in each:

- **Report A.** Before any scores were seen, the study plan declared a family of two comparisons, each music condition against Silence: Instrumental music minus Silence and Music with lyrics minus Silence, with 95% simultaneous coverage for the pair.
- **Report B.** No comparison was declared in advance. After inspecting the three means, an analyst reports the largest pairwise difference, Music with lyrics minus Silence.

The Bonferroni critical value for two comparisons is 2.373417201, and Report A's 95% simultaneous intervals are $[-4.7253, 0.7253]$ for Instrumental music minus Silence and $[-6.7253, -1.2747]$ for Music with lyrics minus Silence. Tukey's 95% all-pairs intervals are $[-4.847, 0.847]$ for Instrumental music minus Silence, $[-6.847, -1.153]$ for Music with lyrics minus Silence and $[-4.847, 0.847]$ for Music with lyrics minus Instrumental music.

1. Before calculating, state what differs between the two reports and what effect the difference has on the family of comparisons that the uncertainty statement must cover. Give your reasons.
2. For Report A, explain how dividing a .05 familywise error allowance between the two intervals and their tails leads to the quantile .9875 behind its critical value. Interpret the two intervals together, and explain why an interval containing zero does not establish equivalence.
3. For Report B, identify the family from which the comparison was selected and the interval warranted for Music with lyrics minus Silence. Compare it with Report A's interval for the same comparison, say which one describes the uncertainty under Report B's history and why, and state what each history lets you conclude about the comparison. A Bonferroni bound over Report B's family uses $t_{1-.05/6;27} = 2.552458806$: calculate its half-width and explain why it exceeds both Report A's half-width, 2.7253, and Tukey's, 2.8470.
4. Report A's two comparisons share the Silence mean. Explain why this creates dependence between the two estimates and why it does not invalidate the Bonferroni bound.

Exercise 5: Read the factorial pattern (Foundational)

Use the complete invented study below. There are four different students per cell, individually allocated to the six combinations.

Feedback	Massed scores	Spaced scores
Rereading	4, 5, 7, 8	4, 5, 7, 8
Answer feedback	6, 7, 9, 10	10, 11, 13, 14
Explanatory feedback	8, 9, 11, 12	16, 17, 19, 20

1. Identify the outcome, the two factors and their levels, the assignment unit, the number of cells and the meaning of balance.
2. Leave practice aside for a moment, so that each feedback condition is one group of eight students. The three feedback means are then 6, 10 and 14, and the two practice means 8 and 12; explain why they equal the equally weighted marginal means here. Analysed this way, the feedback groups have SDs 1.6903, 2.7255 and 4.5981, a within-group mean square of 10.4762 on 21 df and $F(2, 21) = 12.2182$, whereas the full two-way model has $MS_E = 3.3333$ on 18 df and feedback $F(2, 18) = 38.4000$. Explain where the extra within-group variation comes from and why the Explanatory group's SD is the largest.
3. The additive prediction for Explanatory-Spaced, the grand mean plus the Explanatory-feedback and Spaced-practice effects, is 16, and the observed cell mean is 18. Explain what the interaction departure of 2 says about spacing under Explanatory feedback, and why the other departures in its row and column must balance it.
4. The Spaced-minus-Massed differences are 0, 4 and 8 points for Rereading, Answer feedback and Explanatory feedback, so Answer's schedule difference minus Rereading's is 4 points. Explain how these values appear in the cell-mean plot, and what the plot would show if the three differences were equal.
5. Explain why the four-cell difference in part 4 is one interaction contrast, while the full three-by-two interaction has two df. Explain why the connected lines do not represent paired students.

Exercise 6: Distinguish joint and selected results (Integrative)

For the full feedback-by-practice model, the ANOVA table gives feedback $F(2, 18) = 38.4$, practice $F(1, 18) = 28.8$ and interaction $F(2, 18) = 9.6$, with p-values below .0001, below .0001 and .0015, and error mean square $MS_E = 60/18 = 10/3$ on 18 df. Assume independent common-variance Normal errors. The critical values in parts 3 and 4 carry extra decimals so that the printed limits can be reproduced by hand.

1. State in words the joint null hypothesis that each of the three rows tests, the order in which to read the rows, and the reason for that order.

2. The schedule marginal means are 8 and 12, but the conditional schedule differences are 0, 4 and 8. Explain what the practice main effect averages and why it does not describe every condition.
3. Answer's 4-point schedule difference has $SE = \sqrt{5/3}$, and with $t_{.975,18} = 2.10092204$ its pointwise 95% interval is $[1.2877, 6.7123]$. State the family limitation of this single-comparison interval.
4. After inspecting the plot, a reader selects Answer's schedule difference minus Rereading's, estimated as 4. Its SE from four independent cell means of size 4 and $MS_E = 10/3$ is 1.8257, and its Scheffé interval, with multiplier $\sqrt{2F_{.95;2,18}} = 2.666292237$, is $[-0.868, 8.868]$. Identify the protected family and explain why it covers a comparison chosen after the plot was inspected.
5. Explain why the interval in part 4 can contain zero when the omnibus interaction test is significant, and why that interaction-only family does not also protect part 3's simple effect.

Exercise 7: Repair a claim about teaching (Integrative)

Repair this claim about the invented study:

Spaced practice improves every student's learning by four points, whatever feedback is provided. The significant interaction proves that Answer feedback benefits more from spacing than Rereading, and the results apply to all students.

All 24 quiz scores are observed. The stipulated design allocates four different students to each cell. The equal-weight practice main effect is 4 points; conditional schedule differences are 0, 4 and 8. The interaction test gives $F(2, 18) = 9.6$, $p = .0015$. The selected Answer-minus-Rereading difference of schedule differences is 4, with a 95% interaction-family Scheffé interval $[-0.868, 8.868]$.

1. Identify at least four words or claims that exceed the supplied statistical and design information.
2. Write a bounded replacement retaining the conditional pattern, the joint interaction result, the selected estimate and interval, the 0–20 quiz scale, and the invented nature of the example.
3. Explain what random allocation could support in a real study and what it would not establish about population sampling or individual treatment effects.
4. Explain how class-level assignment, repeated observations or missing outcomes would affect the information needed for the analysis. Do not invent losses in the present dataset. State why four observations per cell limit diagnostic reassurance.

Exercise 8: Match the analysis to the design (Integrative)

The Silence scores are 8, 9, 9, 11, 12, 13, 13, 14, 15, 16 and the Music-with-lyrics scores are 4, 5, 6, 7, 8, 8, 9, 10, 11, 12. They belong to twenty different students. The pooled within-group variance is 7. For these matched classical calculations, the two-sided p-value is .0033.

1. Silence minus Music with lyrics is 4 points, with $SE = \sqrt{7(1/10 + 1/10)} = 1.1832$ and pooled $t = 3.3806$ on 18 df, and a one-way ANOVA of the same two groups gives $F(1, 18) = 11.4286$. Explain why F equals t^2 here and why the two p-values agree, and say what F loses that t and the signed difference keep.
2. Suppose instead that each of ten students read one passage in silence and a comparable passage with lyrics, in a random order. Explain why a paired analysis would use within-student differences and why the independent-group calculation in part 1 would no longer be the appropriate starting point. Name the dependence information needed to calculate its SE.
3. Explain how assigning one class to each condition within each school creates a blocked comparison. Identify the comparison units in Chapter 3's paired-school example (Section 2.3) and distinguish an allocation block from a sampling stratum.
4. A researcher proposes measuring a fractions pretest before assigning feedback. Explain what baseline adjustment adds, why an adjusted comparison needs a specified common value or averaging distribution, and why adjustment alone does not establish a causal interpretation.

Appendix A: Deriving the Matched Pooled-t Identity

Section 2.4 states this identity and its conditions; the derivation shows where it comes from. For two independent groups, let $\widehat{\Delta\mu} = \bar{y}_1 - \bar{y}_2$ and $N = n_1 + n_2$. Their grand mean is

$$\bar{y} = \frac{n_1\bar{y}_1 + n_2\bar{y}_2}{N}.$$

Subtracting it from each group mean gives

$$\bar{y}_1 - \bar{y} = \frac{n_2}{N}\widehat{\Delta\mu}, \quad \bar{y}_2 - \bar{y} = -\frac{n_1}{N}\widehat{\Delta\mu}.$$

Consequently,

$$\begin{aligned} SS_B &= n_1 \left(\frac{n_2}{N}\widehat{\Delta\mu} \right)^2 + n_2 \left(-\frac{n_1}{N}\widehat{\Delta\mu} \right)^2 \\ &= \frac{n_1 n_2}{N} \widehat{\Delta\mu}^2. \end{aligned}$$

There is one numerator df, so $MS_B = SS_B$. Define the pooled variance $s_p^2 = SS_W/(N - 2)$, which is Chapter 3's pooled variance and equals MS_W when $J = 2$. Because $n_1 n_2 / N = 1/(1/n_1 + 1/n_2)$,

$$F = \frac{SS_B}{s_p^2} = \frac{\widehat{\Delta\mu}^2}{s_p^2(1/n_1 + 1/n_2)} = t^2,$$

where $t = \widehat{\Delta\mu} / \{s_p \sqrt{1/n_1 + 1/n_2}\}$ is the equal-variance t statistic.

The equality is algebraic for these matched definitions; the equal p-values add the test conditions and a two-sided t alternative. It connects the one-way F test to the equal-variance t test of Chapter 3, Section 5.3. The paired-samples t test (Chapter 3, Section 5.2) and Welch's t test (Chapter 3, Section 5.4) have their own SEs and df, so this algebra supplies neither; Welch's several-group counterpart is the Welch one-way ANOVA of Section 5.2.

Appendix B: Weighting and Effect-Size Reference

B.1 Unequal-cell weights

In the chapter's balanced data, equal-cell and student-weighted marginal means coincide. For unequal cell counts, define them separately:

$$\bar{y}_{j\cdot}^{\text{students}} = \frac{\sum_k n_{jk} \bar{y}_{jk}}{\sum_k n_{jk}}, \quad \bar{y}_{j\cdot}^{\text{equal cells}} = \frac{1}{K} \sum_k \bar{y}_{jk}.$$

The first describes the mixture of students observed in that feedback group. The second gives each of the K practice conditions the same weight. Neither is automatically the right target for every question.

For an illustration, retain Answer's cell means 8 and 12 but imagine counts of 2 Massed and 6 Spaced students. The equally weighted mean is $(8 + 12)/2 = 10$; the student-weighted mean is $(2 \times 8 + 6 \times 12)/8 = 11$. These are hypothetical weights, not changes to the actual four-per-cell dataset. A comparison of feedback groups using different schedule mixtures can combine a feedback comparison with a weighting difference. Adjusted means in later chapters make their common profile or averaging distribution explicit for this reason.

B.2 Effect-size denominators

Section 2.4 used one omnibus summary, eta squared. Software reports several related summaries whose denominators differ, so each needs its name and its denominator before it is interpreted.

Table 20*Four effect-size summaries and their denominators.*

Summary	Formula	Denominator	Value here
Eta squared, one-way	SS_B/SS_T	All observed variation	.3101
Omega squared, one-way	$\frac{SS_B - (J - 1)MS_W}{SS_T + MS_W}$	All observed variation plus one error mean square; the numerator first removes the variation expected from noise	.2525
Eta squared for a factorial term	SS_{term}/SS_T	All observed variation, including the other factors	Interaction .1345
Partial eta squared	$\frac{SS_{\text{term}}}{SS_{\text{term}} + SS_E}$	The term and the error only	Interaction .5161

Here $\eta^2 = 80/258 = .3101$, $\omega^2 = \{80 - 2(178/27)\}/(258 + 178/27) = .2525$, and the interaction's two summaries are $64/476 = .1345$ and $64/(64 + 60) = .5161$. Omega squared subtracts the between-group variation that group means show even when the model means are equal; it rests on the common-variance model, can fall below zero in a small sample, and like eta squared says nothing about causation. The two factorial rows show why the name matters: the same interaction is .5161 of its partial denominator but .1345 of all the variation, because the partial version leaves the feedback and practice components out. In one-way ANOVA the two eta squared versions agree, since the total holds only the factor and error. The raw mean differences remain necessary for understanding the outcome scale.

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